Define $l = \sqrt{\frac{\hbar}{m\omega}}$ as the oscillator's length scale

$$\dim\left(\frac{\hbar}{m\omega}\right) = \frac{J \cdot s}{kg \cdot \frac{1}{s}} = \frac{\cancel{kg} \cdot \frac{m}{\cancel{s^2}} \cdot m \cdot \cancel{s}}{\cancel{kg}/\cancel{s}} = m^2$$

$$\Rightarrow \dim(l) = \dim\sqrt{\frac{\hbar}{m\omega}} = \underline{\text{meters}} \quad \text{ok.}$$

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)$$

$$\hat{a}^+ = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)$$

$$m\omega = \frac{\hbar}{l^2} \Rightarrow \frac{1}{m\omega} = \frac{l^2}{\hbar} \qquad \sqrt{m\omega\hbar} = \sqrt{\frac{\hbar^2}{l^2}} = \frac{\hbar}{l}$$

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{l} + i\frac{l}{\hbar}\hat{p}\right)$$

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{l} - i\frac{l}{\hbar}\hat{p}\right)$$

$$\mathrm{Dim}\left(\frac{\hbar}{l}\right) = \frac{J \cdot s}{m} = \frac{kg \cdot m/\cancel{s^2} \cdot \cancel{m} \cdot \cancel{s}}{\cancel{m}} = kg \cdot \frac{m}{s} = \mathrm{Dim}(\hat{p}).$$

So that finally
$$\begin{cases} \hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{l} + i\frac{\hat{p}}{\hbar/l}\right) \\ \hat{a}^+ = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{l} - i\frac{\hat{p}}{\hbar/l}\right) \end{cases}$$

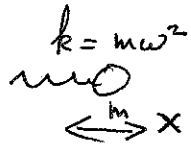
$$\begin{cases} \hat{x} = \frac{l}{\sqrt{2}}\left(\hat{a}^+ + \hat{a}\right) \\ \hat{p} = \frac{\hbar}{l}\frac{i}{\sqrt{2}}\left(\hat{a}^+ - \hat{a}\right) \end{cases}$$

Then $\psi_n(x) = \langle x|n\rangle = \langle x| \frac{(\hat{a}^+)^n}{\sqrt{n!}} |0\rangle =$

$$= \frac{1}{\sqrt{n!}} \langle x| \left[\frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{\ell} - i\frac{\ell}{\hbar}\hat{p}\right)\right]^n |0\rangle = \left(\hat{p} = \frac{\hbar}{i}\frac{d}{dx}\right) =$$

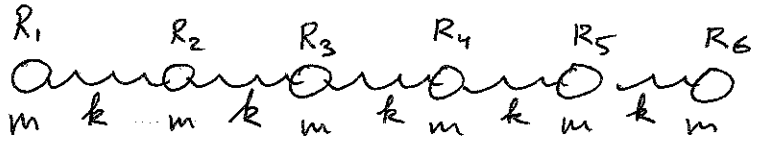
$$= \frac{1}{\sqrt{2^n n!}} \langle x| \left(\frac{x}{\ell} - \ell\frac{d}{dx}\right)^n |0\rangle$$

Harmonic oscillator



$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2 \longrightarrow \hbar\omega\left(a^+ a + \frac{1}{2}\right)$$

Let us now consider the <u>chain of coupled oscillators</u>



$R_j = R_j^0 + x_j$

equilibrium; deviation

→ Classical energy $K.E. = \sum_{j=1}^{N} \frac{p_j^2}{2m}$

$$P.E. = \frac{1}{2} k \left(x_j - x_{j-1}\right)^2 = \frac{1}{2} m\omega^2 \sum_{j=1}^{N} \left(x_j - x_{j-1}\right)^2$$

$$\Rightarrow H = \sum_{j=1}^{N} \left(\frac{p_j^2}{2m} + \frac{1}{2} k \left(x_j - x_{j-1}\right)^2\right)$$

only deviations from equilibrium <u>cost</u> energy. Also, uniform deviation does not cost anything.

→ the Lagrangian

$$L = KE - PE$$

$$L = \sum_{j=1}^{N} \left[\frac{p_j^2}{2m} - \frac{1}{2} k \left(x_j - x_{j-1}\right)^2\right]$$

Lagrange equation of motion $(p = m\dot{x})$

$$\frac{\partial L}{\partial x_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} = 0$$

$$\frac{\partial L}{\partial x_i} = -\frac{1}{2} k \frac{\partial}{\partial x_i} \sum_{j=1}^{N} (x_j - x_{j-1})^2 =$$

$$= -\frac{1}{2} k \frac{\partial}{\partial x_i} \left[(x_i - x_{i-1})^2 + (x_{i+1} - x_i)^2 \right] =$$

$$= -\frac{1}{2} k \frac{\partial}{\partial x_i} \left[x_i^2 - 2 x_i x_{i-1} + x_i^2 - 2 x_i x_{i+1} + \dots \right]$$

$$= -\frac{1}{2} k \left[2 x_i + 2 x_i - 2 x_{i-1} - 2 x_{i+1} \right] =$$

$$= -k \left(2 x_i - x_{i-1} - x_{i+1} \right)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} = \frac{d}{dt}\frac{\partial}{\partial \dot{x}_i}\left(\frac{m}{2}\dot{x}_j^2\right) = \frac{d}{dt} m \dot{x}_i = m \ddot{x}_i$$

$$\Rightarrow \quad m \ddot{x}_i = -k \left(2 x_i - x_{i-1} - x_{i+1} \right) = k \left(x_{i+1} + x_{i-1} - 2 x_i \right)$$

(classical equation of motion)

We now exploit periodicity $\quad x_j = x_{j+N}$

$$\cancel{x_j = \int_{-\pi}^{\pi} \frac{da}{2\pi} e^{iqj} x_q}$$

(NOTE that realistically $N \sim 10^{23}$ atoms!)

~~The [illegible] follow from~~

We use discrete Fourier transform $\left(a = R_j^0 - R_{j-1}^0 = \text{lattice spacing}\right)$

$$X_j = \frac{1}{\sqrt{N}} \sum_q X_q \, e^{iqja}$$

from $X_j = X_{j+N}$

$$\frac{1}{\sqrt{N}} \sum_q X_q \, e^{iqja} = \frac{1}{\sqrt{N}} \sum_q X_q \, e^{iq(j+N)a}$$

$$\Rightarrow \quad qNa = 2\pi n \quad , \; n = \text{Integer}$$

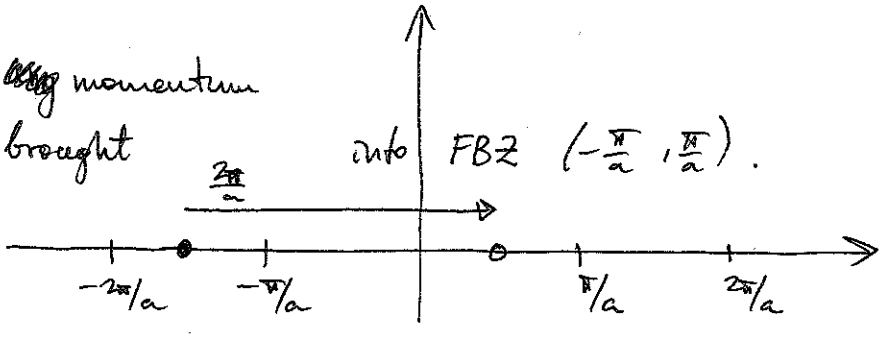
$$q = \frac{2\pi n}{Na} \qquad n = 0, \pm 1, \pm 2, \ldots \pm N$$

$$\Rightarrow \boxed{-\frac{\pi}{a} \le q \le \frac{\pi}{a}}$$ the First Brillouin Zone

This is because $e^{iqja} = e^{i(q + \frac{2\pi}{a})ja} = e^{iqja} e^{i(2\pi j)}$

but j = integer $\Rightarrow$ $\underline{e^{i2\pi j} = 1}$.

arbitrary ~~wavy~~ momentum can be brought into FBZ $\left(-\frac{\pi}{a}, \frac{\pi}{a}\right)$.



Thus $$\frac{1}{\sqrt{N}} \sum_q m \ddot{X}_q \, e^{iqja} = \frac{1}{\sqrt{N}} \sum_q k \, X_q \left[e^{iq(j+1)a} + e^{iq(j-1)a} - 2 \, e^{iqja} \right] =$$

$$= \frac{1}{\sqrt{N}} \sum_q k \, X_q \, e^{iqja} \left(2 \cos qa - 2\right)$$

Seek stationary solution $X_q(t) = X_q\, e^{-iEt/\hbar}$

$$\dot{X} = -i\frac{E}{\hbar} X$$
$$\ddot{X} = -\frac{E^2}{\hbar^2} X$$

$\Rightarrow$ finally

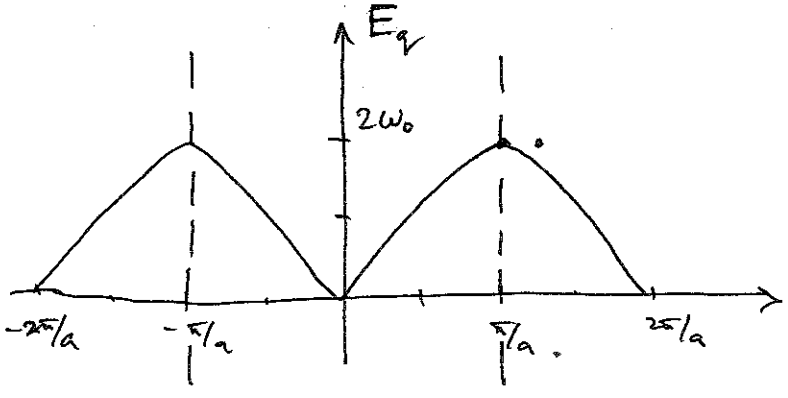
$$\frac{1}{\sqrt{N}} \sum_{q \in BZ} \left[-m\frac{E^2}{\hbar^2} + 2k(1-\cos qa) \right] X_q\, e^{iqja} = 0$$

$$\Rightarrow E_q^2 = \hbar^2 \frac{2k}{m}(1-\cos qa) = \hbar^2 \frac{4k}{m} \sin^2\frac{qa}{2}$$

$$\cos qa = \cos^2\frac{qa}{2} - \sin^2\frac{qa}{2} = 2\cos^2\frac{qa}{2} - 1 = 1 - 2\sin^2\frac{qa}{2}$$

$$E_q = \hbar 2\sqrt{\frac{k}{m}}\left|\sin\frac{qa}{2}\right| \qquad \left(\text{or } k = m\omega_0^2\right)$$

$$E_q = \hbar 2\omega_0 \left|\sin\frac{qa}{2}\right| \Rightarrow \omega_q = 2\omega_0\left|\sin\frac{qa}{2}\right|.$$



Lattice oscillations = phonons:

For every momentum $q \in (-\pi/a, \pi/a)$ there is energy $E_q = 2\omega_0\left|\sin\frac{qa}{2}\right|$ that corresponds to propagating elastic wave of deformation.

For $q \to 0$ $\quad E_q \to 2\omega_0 \frac{qa}{2} = \omega_0 a q = \sqrt{\frac{k}{m}}\, a q$

describes propagating SOUND wave; $c = a\sqrt{\frac{k}{m}}$

<u>Quantum</u> lattice of ions ("matress")

$$\hat{H} = \sum_{j=1}^{N} \left[\frac{\hat{p}_j^2}{2m} + \frac{1}{2} k \left(\hat{x}_j - \hat{x}_{j-1} \right)^2 \right]$$

but $[\hat{x}_j, \hat{p}_m] = i\hbar\, \delta_{jm}$

(position and momentum do not commute only if they describe motion of the same ion #j.)

FT
$$\begin{cases} \hat{X}_j = \frac{1}{\sqrt{N}} \sum_q \hat{X}_q \, e^{iqja} \\ \hat{P}_j = \frac{1}{\sqrt{N}} \sum_q \hat{P}_q \, e^{iqja} \end{cases}$$

Inverse FT
$$\begin{cases} \hat{X}_q = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \hat{X}_j \, e^{-iqja} \\ \hat{P}_q = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \hat{P}_j \, e^{-iqja} \end{cases}$$

commutator
$$[\hat{X}_q, \hat{P}_k] = \frac{1}{N} \sum_{j=1}^{N} \sum_{m=1}^{N} \left[\hat{X}_j \, e^{-iqja}, \hat{P}_m \, e^{-ikma} \right] =$$

$$= \frac{1}{N} \sum_{jm} \underbrace{[\hat{X}_j, \hat{P}_m]}_{i\hbar\, \delta_{jm}} e^{-ia(qj+km)} =$$

$$= i\hbar \underbrace{\frac{1}{N} \sum_m e^{-i(q+k)ma}}_{\delta_{q+k,0}} = i\hbar\, \delta_{q+k,0}$$

$$\sum_{j=1}^{N} \hat{p}_j^2 = \frac{1}{N} \sum_{j=1}^{N} \sum_{qq'} \hat{p}_q \hat{p}_{q'} e^{iqja + iq'ja} =$$

$$= \left(\frac{1}{N} \sum_{j=1}^{N} e^{ij(q+q')a} = \delta_{q+q',0} \right) = \sum_q \hat{p}_q \hat{p}_{-q}$$

$$\sum_{j=1}^{N} (x_j - x_{j-1})^2 = \frac{1}{N} \sum_{qq'} \sum_j \hat{x}_q \left(e^{iqja} - e^{iq(j-1)a} \right) \cdot$$

$$\cdot \hat{x}_{q'} \left(e^{iq'ja} - e^{iq'(j-1)a} \right) =$$

$$= \frac{1}{N} \sum_{qq'} \hat{x}_q \hat{x}_{q'} \sum_{j=1}^{N} e^{ijaq} \left(1 - e^{-iqa}\right) e^{ijaq'} \left(1 - e^{-iq'a}\right)$$

$$= \left(\frac{1}{N} \sum_{j=1}^{N} e^{ia(q+q')j} = \cancel{\delta(q+q')} \; \delta_{q+q',0} \right) = \sum_q \hat{x}_q \hat{x}_{-q} \underbrace{\left(1 - e^{-iqa}\right)\left(1 - e^{iqa}\right)}_{1+1-2\cos qa}$$

$$= \sum_q 2(1 - \cos qa) \hat{x}_q \hat{x}_{-q} = \sum_q 4 \sin^2\left(\frac{qa}{2}\right) \hat{x}_q \hat{x}_{-q}$$

$$\Rightarrow \hat{H} = \sum_{q \in BZ} \left\{ \frac{1}{2m} \hat{p}_q \hat{p}_{-q} + k \sin^2\left(\frac{qa}{2}\right) \hat{x}_q \hat{x}_{-q} \right\}$$

$$k = m\omega_0^2$$

$$E_q^2 = 4\omega_0^2 \sin^2 \frac{qa}{2}$$

$$= \sum_q \left\{ \frac{1}{2m} \hat{p}_q \hat{p}_{-q} + \frac{1}{2} m E_q^2 \hat{x}_q \hat{x}_{-q} \right\}$$

$$E_q = 2\omega_0 \left| \sin \frac{qa}{2} \right| = E_{-q}$$

$\Rightarrow \hat{H}$ is (almost) collection of harmonic oscillators at momenta q.

(the difference is $\hat{p}_q \hat{p}_{-q}$ & $\hat{x}_q \hat{x}_{-q}$ instead of $\hat{p}_q^2$ & $\hat{x}_q^2$)

We now introduce

$$\left.\begin{aligned} \hat{x}_q &= \sqrt{\frac{\hbar}{m E_q}}\ \frac{b_q + b^+_{-q}}{\sqrt{2}} \\ \hat{p}_q &= \sqrt{\hbar\, m E_q}\ \frac{b_q - b^+_{-q}}{i\sqrt{2}} \end{aligned}\right\}$$

check below that all comm. relations are satisfied.

$$\hat{H} = \sum_q \left\{ \frac{\hbar m E_q}{2m} \frac{-1}{2} \left(b_q - b^+_{-q}\right)\left(b_{-q} - b^+_q\right) + \frac{1}{2} m E_q^2 \frac{\hbar}{m E_q} \cdot \frac{1}{2} \cdot \right.$$

$$\left. \cdot \left(b_q + b^+_{-q}\right)\left(b_{-q} + b^+_q\right) \right\} =$$

$$= \sum_q \left\{ \frac{-1}{4} \hbar E_q \left(\underline{b_q b_{-q}} - b_q b^+_q - b^+_{-q} b_{-q} + \underline{\underline{b^+_{-q} b^+_q}} \right) \right.$$

$$\left. + \frac{1}{4} \hbar E_q \left(\underline{b_q b_{-q}} + b_q b^+_q + b^+_{-q} b_{-q} + \underline{\underline{b^+_{-q} b^+_q}} \right) \right\}$$

$$= \sum_q \frac{1}{2} \hbar E_q \left(b_q b^+_q + b^+_{-q} b_{-q} \right) =$$

$$= \sum_q \frac{1}{2} \hbar E_q \underset{q \to -q}{b^+_{-q} b_{-q}} + \sum_q \frac{1}{2} \hbar E_q \left(1 + b^+_q b_q\right) =$$

$$= \sum_q \frac{1}{2} \hbar E_q\, b^+_q b_q + \sum_q \frac{1}{2} \hbar E_q \left(b^+_q b_q + 1\right)$$

$$= \sum_q \hbar E_q \left(b^+_q b_q + \frac{1}{2} \right)$$

$\Rightarrow$ It is just a collection of harmonic oscillators: one for each momentum $q = \frac{\pi n}{Na}$ ($n = 0, \pm 1, \pm 2 \ldots$ inside the first Brillouin zone. <u>PHONONS</u>

Check commutation relations

$$\hat{x}_q + \frac{i\hat{p}_q}{mE_q} = \sqrt{\frac{\hbar}{mE_q}} \underbrace{\left(\frac{b_q + b_{-q}^+}{\sqrt{2}} + \frac{b_q - b_{-q}^+}{\sqrt{2}} \right)}_{\sqrt{2}\, b_q}$$

$$\Rightarrow b_q = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_q + \frac{i}{mE_q} \hat{p}_q \right)$$

$$\cancel{\Rightarrow b_q^+ = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_q - \frac{i}{mE_q} \hat{p}_q \right)}$$

$$\cancel{\Rightarrow b_{-q}^+ = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_{-q} - \frac{i}{mE_q} \hat{p}_{-q} \right)}$$

This can be checked as

$$\hat{x}_{+q} - \frac{i}{mE_q} \hat{p}_{+q} = \sqrt{\frac{\hbar}{mE_q}} \underbrace{\left(\frac{b_{+q} + b_{-q}^+}{\sqrt{2}} - \frac{b_{+q} - b_{-q}^+}{\sqrt{2}} \right)}_{\sqrt{2}\, b_{-q}^+}$$

$$\Rightarrow b_{-q}^+ = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_q - \frac{i}{mE_q} \hat{p}_q \right)$$

thus $b_q^+ = \sqrt{\frac{mE_q}{2\hbar}} \left(\hat{x}_{-q} - \frac{i}{mE_q} \hat{p}_{-q} \right)$

Note that $\left[\hat{b}_q, \hat{b}_k^+ \right] = \sqrt{\frac{mE_q}{2\hbar}} \sqrt{\frac{mE_k}{2\hbar}} \left[\hat{x}_q + \frac{i}{mE_q} \hat{p}_q, \hat{x}_{-k} - \frac{i}{mE_k} \hat{p}_{-k} \right]$

$$= \frac{m}{2\hbar} \sqrt{E_q E_k} \left(\frac{-i}{mE_k} \underbrace{\left[\hat{x}_q, \hat{p}_{-k} \right]}_{i\hbar\, \delta_{q-k,0}} + \frac{i}{mE_q} \left[\hat{p}_q, \hat{x}_{-k} \right] \right)$$

$$-i\hbar\, \delta_{q-k,0}$$

$$= \frac{m}{2\hbar} E_k \frac{2\hbar}{mE_k} \delta_{q-k,0} = \delta_{q-k,0}$$

This is how it must be because

$$\hat{x}_q = \frac{1}{\sqrt{N}} \sum_j \hat{x}_j e^{-iqja}$$

but $$\left(\hat{x}_q\right)^+ = \frac{1}{\sqrt{N}} \sum_j \underset{\hat{x}_j}{\underset{\parallel}{\hat{x}_j^+}} e^{iqja} = \frac{1}{\sqrt{N}} \sum_j \hat{x}_j e^{-i(-q)ja} = \hat{x}_{-q}$$

The same for momentum: $\hat{p}_q^+ = \hat{p}_{-q}$

Thus, while in position space $\hat{x}$ and $\hat{p}$ are Hermitian (self-adjoint) operators, in momentum space they are not!

Hamiltonian of phonons:

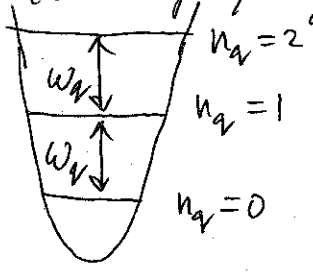
$$\hat{H}_{ph} = \sum_q \hbar\omega_q \left(b_q^+ b_q + \frac{1}{2} \right) =$$

$$= \sum_q \hbar\omega_q \left(\hat{n}_q + \frac{1}{2} \right) = \sum_q \hat{H}_{SHO}(q)$$

hamiltonian of q-th simple harmonic oscillator.

where $\hat{n}_q = b_q^+ b_q$ is the occupation number operator which counts number of phonons in q-th harmonic oscillator.

* q-th oscillator: frequency $\omega_q = 2\omega_0 \left| \sin \frac{qa}{2} \right|$.



Notice that number of phonons n_q with momentum q ~~is~~ the same as specifying that the oscillator is in n_q-th excited state.

The energy of the state with $\underline{n_q}$ phonons is $\underline{\hbar\omega_q \left(n_q + \frac{1}{2} \right)}$.

Modes with different wavenumbers q are independent from each other $\Rightarrow$ partition function of the lattice

$$Z = \prod_q Z_q$$

is the product of partition functions describing oscillators with frequency ω_q.

Thus
$$Z_q = \sum_{n_q=0}^{\infty} e^{-\frac{\hbar\omega_q}{\tau}\left(n_q+\frac{1}{2}\right)} = \frac{e^{-\frac{\hbar\omega_q}{2\tau}}}{1-e^{-\frac{\hbar\omega_q}{\tau}}} = \frac{1}{2\sinh\left(\frac{\hbar\omega_q}{2\tau}\right)}$$

Number of phonons in q-th oscillator = occupation number of q-th oscillator

$$\langle n_q \rangle = \frac{\sum_{n_q=0}^{\infty} n_q\, e^{-\frac{\hbar\omega_q}{\tau}\left(n_q+\frac{1}{2}\right)}}{Z_q} =$$

$$= \frac{e^{-\frac{\hbar\omega_q}{2\tau}}}{Z_q} \sum_{n_q=0}^{\infty} n_q\, e^{-\frac{\hbar\omega_q}{\tau} n_q} = \frac{e^{-\frac{\hbar\omega_q}{2\tau}}}{Z_q}\left(-\frac{\partial}{\partial\left(\frac{\hbar\omega_q}{\tau}\right)} \sum_{n_q} e^{-\frac{\hbar\omega_q}{\tau} n_q}\right)$$

$$= \frac{e^{-\frac{\hbar\omega_q}{2\tau}}}{Z_q}\left(-\frac{\partial}{\partial\left(\frac{\hbar\omega_q}{\tau}\right)} \frac{1}{1-e^{-\frac{\hbar\omega_q}{\tau}}}\right)$$

But $\frac{\partial}{\partial x}\frac{1}{1-e^{-x}} = \frac{-e^{-x}}{(1-e^{-x})^2}$

$$\Rightarrow \langle n_q \rangle = \left(1-e^{-\frac{\hbar\omega_q}{\tau}}\right)\frac{e^{-\frac{\hbar\omega_q}{\tau}}}{\left(1-e^{-\frac{\hbar\omega_q}{\tau}}\right)^2} = \frac{e^{-\frac{\hbar\omega_q}{\tau}}}{1-e^{-\frac{\hbar\omega_q}{\tau}}} = \frac{1}{e^{\frac{\hbar\omega_q}{\tau}}-1}$$

Planck distribution function

Thermal energy of q-th oscillator

$$U_q = \langle \varepsilon_q \rangle = \hbar\omega_q \langle n_q \rangle = \frac{\hbar\omega_q}{e^{\frac{\hbar\omega_q}{\tau}} - 1}$$

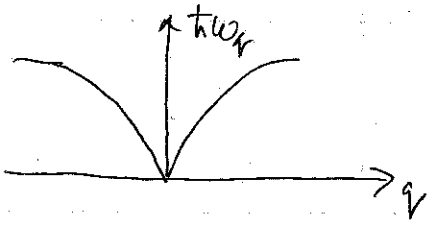
$\Rightarrow$ thermal energy of phonons gas is obtained by summing over all q from the Brillouin zone

$$U = \sum_q U_q = \sum_q \frac{\hbar\omega_q}{e^{\frac{\hbar\omega_q}{\tau}} - 1}$$

In our one-dimensional chain $\hbar\omega_q = 2\omega_0 \left|\sin\frac{qa}{2}\right|$ $\left(\omega_0 = \sqrt{\frac{K}{m}}\right)$ and $q \to q_n = \frac{2\pi n}{Na}$, with N independent q values.

$\boxed{D=3}$ In three-dimensional crystals $\hbar\omega_q$ has similar structure:

$\hbar\omega_q = vq$ at small $q = |\vec{q}| = \sqrt{q_x^2 + q_y^2 + q_z^2}$



and reaches maximum at the boundaries of 3-dimensional Brillouin zone.

At low temperature τ phonons with energies $\hbar\omega_q \sim \tau$ are most important $\Rightarrow$ we can approximate

$\omega_q = vq$, where $v \approx \sqrt{\frac{K}{m}}$ is the speed of sound.

Each atom has 3 independent directions to oscillate

$\Rightarrow$ there $3N$ independent oscillators in that case.

This is described by having N oscillators with 3 independent polarizations: $\begin{cases} 2 \text{ transverse} \\ 1 \text{ longitudinal.} \end{cases}$

$$N \cdot 3 = 3N.$$

Evaluate sum over (q) by converting it into integral. For example: consider sum over q_x

$$\sum_{q_x} f(q_x) = \sum_{n_x=-\frac{N}{2}}^{N/2} f\left(\frac{2\pi}{L_x} n_x\right) = \int_{-\frac{N}{2}}^{\frac{N}{2}} dn_x \, f\left(\frac{2\pi}{L_x} n_x\right)$$

$$q_x = \frac{2\pi n_x}{Na} = \frac{2\pi}{L_x} n_x$$

but $dn_x = \frac{L_x}{2\pi} dq_x \Rightarrow$

$$\int_{-N/2}^{N/2} dn_x \, f\left(\frac{2\pi n_x}{L_x}\right) = \frac{L_x}{2\pi} \int_{-\pi/a}^{\pi/a} dq_x \, f(q_x)$$

The factor $\frac{L_x}{2\pi} \equiv \frac{dq_x}{(\Delta q_x)} = \frac{dq_x}{(2\pi/L_x)}$ is the density of states for considered 1d example.

It counts number of modes in interval dq_x by dividing the interval by the minimal step $\Delta q_x = \frac{2\pi}{L_x}$.

$\boxed{D=3}$ Then $\sum_{\vec{q}} f(q) = \sum_{q_x, q_y, q_z} f(q_x, q_y, q_z) =$

$$= \underbrace{\frac{L_x}{2\pi} \frac{L_y}{2\pi} \frac{L_z}{2\pi}}_{\frac{V}{(2\pi)^3} = \text{3-dimensional density of states}} \int_{-\pi/a}^{\pi/a} dq_x \, dq_y \, dq_z \, f(q_x, q_y, q_z).$$

For spherically isotropic crystal $dq_x dq_y dq_z = 4\pi q^2 dq$

$$\Rightarrow \sum_q f(q) = \frac{V}{(2\pi)^3} 4\pi \int_0^{q_D} dq \, q^2 f(q).$$

(Everywhere above single polarization is assumed!)

The upper limit of integration, q_d, is related to Debye frequency $\omega_d = v q_d$.
We determine it by requiring

$$\sum_q 1 = 3N = \text{total number of } q\text{-modes,}$$

including 3 polarizations.

Thus $$(3)\cdot\frac{4\pi V}{(2\pi)^3}\cdot\underbrace{\int_0^{q_d} dq\, q^2}_{\frac{1}{3}q_d^3} = 3N$$

$$\frac{V}{2\pi^2}\, q_d^3 = 3N \Rightarrow q_d = \left(\frac{6\pi^2 N}{V}\right)^{1/3}.$$

$\Rightarrow$ Debye frequency $\omega_d = v q_d$.

Then finally

$$U = \sum_q \frac{\hbar\omega_q}{e^{\frac{\hbar\omega_q}{\tau}}-1} = (3)\,\frac{V}{2\pi^2}\int_0^{q_d} dq\, q^2\,\frac{\hbar v q}{e^{\frac{\hbar v q}{\tau}}-1}.$$

Introduce dimensionless $x = \frac{\hbar v q}{\tau} \Rightarrow dq = \frac{\tau}{\hbar v}\,dx$

$$U = \frac{3V}{2\pi^2}\left(\frac{\tau}{\hbar v}\right)^3\int_0^{x_d} dx\, x^2\,\frac{x\tau}{e^x-1} = \frac{3V}{2\pi^2}\,\frac{\tau^4}{(\hbar v)^3}\int_0^{\theta/\tau}\frac{dx\, x^3}{e^x-1}.$$

$$x_d = \frac{\hbar v q_d}{\tau} = \frac{\hbar v}{\tau}\left(\frac{6\pi^2 N}{V}\right)^{1/3} \equiv \frac{\theta}{\tau}$$

$$\theta = \hbar v\left(\frac{6\pi^2 N}{V}\right)^{1/3} = \text{Debye temperature.}$$

$$U = \frac{3V}{2\pi^2} \frac{\tau^4}{(\hbar v)^3} \int_0^{\theta/\tau} dx \, \frac{x^3}{e^x - 1}$$

(1) Low-τ limit : $\frac{\theta}{\tau} \gg 1 \Rightarrow$ approximately

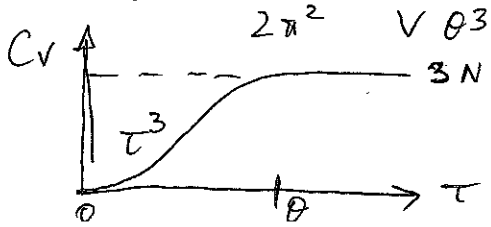
$$U = \frac{3V}{2\pi^2} \frac{\tau^4}{(\hbar v)^3} \underbrace{\int_0^{\infty} dx \, \frac{x^3}{e^x - 1}}_{\frac{\pi^4}{15}}$$

Note that $(\hbar v)^3 = \left(\frac{\theta}{\left(\frac{6\pi^2 N}{V}\right)^{1/3}} \right)^3 = \frac{V \theta^3}{6\pi^2 N}$

$$\Rightarrow U = \frac{3V}{2\pi^2} \frac{\pi^4}{15} \frac{6\pi^2 N}{V\theta^3} \tau^4 = \frac{3\pi^4 N}{5\theta^3} \tau^4$$

$$\Rightarrow C_V = \left(\frac{\partial U}{\partial \tau}\right)_V = \frac{12\pi^4 N}{5} \left(\frac{\tau}{\theta}\right)^3 .$$

(2) High-τ limit : $\frac{\theta}{\tau} \ll 1 \Rightarrow$ expand e^x in denominator in small x

$$U = \frac{3V}{2\pi^2} \frac{\tau^4}{\left(\frac{V\theta^3}{6\pi^2 N}\right)} \underbrace{\int_0^{\theta/\tau} dx \, \frac{x^3}{(1+x)-1}}_{\int_0^{\theta/\tau} dx \, x^2 = \frac{1}{3}\left(\frac{\theta}{\tau}\right)^3}$$

$$\Rightarrow U = \frac{3V}{2\pi^2} \frac{6\pi^2 N}{V\theta^3} \tau^4 \frac{1}{3} \frac{\theta^3}{\tau^3} = 3N\tau \quad \text{(Dulong-Petit limit)}$$